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Hearing in reindeer (*Rangifer tarandus*)

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Abstract The audiogram of two yearling male reindeer (*Rangifer tarandus tarandus*) were determined using a conditioned suppression/avoidance procedure. During testing, the animal was drinking from a metal bowl while pure tone signals were played at random intervals and followed by an electric shock in the bowl. By breaking contact with the bowl at sound signals, the animal avoided the shock. The animals detected sounds at intensities of 60 dB or less from 70 Hz to 38 kHz. The frequency range of best sensitivity was relatively flat from 1 kHz to 16 kHz, with a best sensitivity of 3 dB at 8 kHz. The hearing ability of reindeer is similar to the hearing ability of other ungulates.

Keywords *Rangifer tarandus tarandus* · Audiogram · Conditioned · Suppression/avoidance · Hearing · Frequency

Introduction

Four species of the deer family (Cervidae) are present in Norway. Wild and semi-domestic reindeer (*Rangifer tarandus tarandus*) inhabit a major part of the alpine region, while moose (*Alces alces*) and roe deer (*Capreolus capreolus*) are located in the boreal forests in continental Norway, leaving the red deer (*Cervus elaphus*) as the dominant deer species in the coastal areas of western Norway. The four species share habitats with humans and are exposed to activities relating to settlement, construction, vehicular traffic and tourism. As noises appear to be intimately connected to human activities, the question of hearing ability in these large herbivores is frequently asked. The hearing ability of

domestic ungulates such as cattle (*Bos taurus*), horse (*Equus caballus*), goat (*Capra hircus*), pig (*Sus scrofa*) and sheep (*Ovis aries*) has already been studied (R.S. Heffner and H.E. Heffner 1983, 1990; Wollack 1963). Although it is possible to extrapolate from domestic species and estimate within broad limits the hearing ability in deer, this approach does not meet either public or scientific requirements. We found it challenging and necessary to establish the reindeer audiogram.

Materials and methods

Animals

Two reindeer males, 14 months old at the start of testing, were used in the experiments. The reindeer were taken from a semi-domestic herd (Vågå Tamreinlag) in southern Norway, and held in captivity from an age of 6 months. They had free access to food, but water was given only during testing. They were held in a shaded outdoor cage (2 m×4 m), and were let loose in a grazing area of 40 m×40 m for a few hours every day. Their water consumption and body weight were measured daily to assure that sufficient amounts of water were given. The animals increased their body weight from 55 kg in May 1999 at the beginning of the experiment to 78 kg and 79 kg, respectively, at the end 8 months later.

Experimental chamber

Testing was conducted in a double-walled sound chamber with outside dimensions of 3.6 m×2.4 m×2.4 m, and a wall thickness of 15 cm. The walls and ceiling were covered with egg-crate foam. The animal stood in a 1 m high and 40 cm wide stall made of welded wire mounted on a metal frame. The front of the stall was 70 cm high with the water bowl located 10 cm in front of it and 75 cm above the floor. The animals were tested with sound presented separately from the front and from behind, with the speaker located at 1 m horizontal distance in front or behind the ears of the animal when it was in drinking position. Contact with the bowl was recorded through closing of an electric circuit with the electrode ends attached to the neck of the animal and to the water bowl. The shock at the end of sound signals was delivered through the same circuit. The 40 W signal lamp flashing simultaneously with the shock was located 60 cm above the front speaker.

Low frequent background noise from the ventilation system of the building, and drinking noise from the animals, were present in the experimental chamber during testing (Table 1). Drinking noise

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Table 1 Background and drinking noise levels measured in octave bands at different frequencies

Frequency (Hz)	Background noise (dB)	Drinking noise (dB)
32	45	—
63	33	45
125	25	46
250	16	47
500	10	49
1 k	9	54
2 k	9	55
4 k	9	54
8 k	10	50
16 k	9	43

arose when the animal was licking water from the metal bowl. It occurred simultaneously with the sound signals, but the intensity was variable, reducing to a minimum between licks. The background noise level was within specifications given by the American National Standards Institute for maximum permissible ambient noise levels allowed in audiometric test rooms (Frank and Williams 1993).

Sound stimulation

Pure tones were produced by a sine-wave generator (Newtronics 200PC), turned on and off with an electronic switch, and delivered through a rise/fall gate (Coulborn S84) to avoid onset and offset clicks. The signal was attenuated with a Feiring Leader LAT-45 and lead by an amplifier (Denon PMA 735R) to a speaker (Cerwin-Vega AL-750, Seas 72 K and Paradigm Atom V2 for frequencies up to 8 kHz; and Motorola 1005, Piezo CH-1 and MG Electronics RBT-100 for frequencies above 8 kHz). Above 8 kHz the signal was high-pass filtered (Norwegian Electronics, model 724).

A sound signal lasted for 3 s and was presented in pulses (0.4 s on, 0.1 s off), with a rise/fall time of 20 ms. The sound pressure level (dB re. 20 μ Pa) was measured with a Brüel and Kjær 2231 sound-level meter, with a Brüel and Kjær 1624-octave filter from 32 Hz to 16 kHz, and with a Brüel and Kjær 1627 infra/ultra sound filter for frequencies below 32 Hz and above 16 kHz, respectively. Three different microphones were used depending on frequency (Brüel and Kjær 4155, 4191 and 4939). The microphones were calibrated with a pistonphone (Brüel and Kjær 4231). The electrical signal going to the speaker and the acoustic signal from the speaker was monitored for distortion and noise on an oscilloscope (Iwatsu SS-5702). To assure reliability of the signal monitoring on the oscilloscope, the sound signals were recorded on a DAT-recorder (Sony TCD-D10 PRO II), and spectrum analyses from these recordings were digitally performed in Canary 1.2.4 sound analysis software. Any possible harmonic content in the signals was lower in intensity than the background noise. Before each test session the microphone was placed in mid-position between the ears of the animal during testing, and the level of the sound signal was adjusted to the wanted intensity. The attenuator was calibrated for linearity. Variation in the sound field around the head of the animal was measured for each frequency by placing the microphone 10 cm above, below, right and left of the normal ear position. The variation was within 5 dB for all frequencies except the mid-frequencies from 1 kHz to 8 kHz. For the mid-frequencies the variation could be up to 15 dB.

Experimental procedure

A behavioral method was used to test the hearing of reindeer. It is based on a conditioned suppression/avoidance response in the animal when exposed to stimuli (H.E. Heffner and R.S. Heffner 1995). The animals were trained to drink from a metal bowl. There was a slow supply of water to the bowl only when the animal made contact. At random time intervals a sound signal was played, followed by a

mild electric shock in the bowl. The animals learned to associate the shock with the signal, and avoided the shock by breaking contact (stop drinking) if the sound signal was detected. A flash of light was presented simultaneously with the shock, signaling that contact with the bowl could safely be resumed when the flash ended.

During training the animals learned to respond correctly more often than by chance after a few sound-shock combinations. After 3–4 weeks with daily training, the performance level of the animals was high enough to begin testing, with false alarm rates of <0.2 and hit rates of >0.9 (H.E. Heffner and R.S. Heffner, 1995; R.S. Heffner and H.E. Heffner 1988). The shock was adjusted individually for each animal to a level that was strong enough to make them break contact with the bowl, but mild enough to avoid fear of drinking. The animals were tested on pure tones in octave steps from 63 Hz to 32 kHz and at 35 kHz and 38 kHz. They were tested in separate sessions (one session per day) with the sound signal played from the front and from the back. This was done because the animals tended to have their pinnae directed backwards when drinking.

A test contained a random number, varying from 0 to 16, of safe trials without any sound followed by a warning trial with a sound signal. Both safe trials and warning trials lasted for 3 s, and were separated from the next trial by a 2 s long inter-trial interval. Animal response was recorded by measuring contact with the water bowl during the final 0.2 s of a trial with the shock delivered at the end of warning trials. The responses were recorded by a computer to determine the number of false alarms (i.e., responses in safe trials), and the number of hits (i.e., responses in warning trials).

This gives a Hit rate = Hits/Warning trials and a False-alarm rate = False/Safe trials. If the animal does not hear the signal, the hit rate and the false-alarm rate will not differ. The probability of getting a hit rate $\geq$ the observed hit rate if the animal does not hear the signal is given by the binomial function,

$$P[X \geq c] = \sum_{x=c}^n \binom{n}{x} p^x (1-p)^{n-x} \quad (1)$$

where c is the number of hits, n is the number of warning trials, and p is the false-alarm rate. A significance level of 0.01 was chosen. When testing the animals on a specific frequency and intensity, a minimum of four warning trials were presented when calculating the significance level. If P was close to 0.01, more than four warning trials would be presented.

The hearing threshold for each frequency was found by gradually decreasing the intensity (usually by 5-dB steps) until the animal no longer responded significantly to the sound. The false-alarm rate was always <0.2. The performance rate on each intensity was calculated: Performance rate = Hit rate – (Hit rate $\times$ False-alarm rate). A performance rate of 0.5 was defined to be the threshold (H.E. Heffner and R.S. Heffner 1995; R.S. Heffner and H.E. Heffner 1988). An average total of 142 safe trials and 32 warning trials were presented during a test session before a threshold could be estimated. An animal was tested on a specific frequency until three thresholds were obtained from separate sessions that differed by 5 dB or less. The means of these three thresholds from separate tests are presented here as the actual thresholds.

The two reindeer were trained and tested in the period from May 1999 to February 2000. They were not accustomed to human contact, and had to be tamed for the 1st month. Once they got used to our presence, it took 3–4 weeks of training before we obtained reliable hearing thresholds from the tests. A slow and gradual improvement in the behavioral performance resulted in lower thresholds at some frequencies in the last months of the test period. As a consequence, each frequency was tested again in January/February. If the results differed from the earlier tests with more than 5 dB, the threshold was determined once more with a complete set of new tests.

Results

Separate average audiograms with sound from the front and from behind were established for the two animals

(Fig. 1). They differed in thresholds within 6 dB, with two exceptions at 4 kHz and 8 kHz with 9 dB and 7 dB, respectively (Table 2). Both animals had their pinnae directed backwards during testing, and the hearing thresholds were higher with sound presented from the front than from behind. This tendency was strong at frequencies above 4 kHz, with a maximum difference of 22 dB at 32 kHz. The differences were also relatively large at the two lowest frequencies of 63 Hz and 125 Hz.

At the 60-dB level, the range of hearing for the two reindeer extends from about 70 Hz to about 38 kHz. There is a broad range of high sensitivity from 500 Hz to 32 kHz, with the very lowest threshold of 3 dB at 8 kHz. At frequencies below and above this range, the threshold increases rapidly.

Discussion

Conditioned suppression/avoidance is a behavioral method which has been used on more than 30 species of mammals in studies on auditory abilities (H.E. Heffner and R.S. Heffner 1995). Since water is most often used as a reward, the animal will produce some kind of noise from drinking during testing. Drinking noise might mask the sound signals, giving higher estimates of

hearing thresholds than the actual ones. In previous studies on hearing in mammals where the conditioned suppression/avoidance method with water as a reward have been used, drinking noise has not been reported to be a problem. Most mammals that have been tested are small (e.g., H.E. Heffner and R.S. Heffner 1985; H.E. Heffner et al. 1994; R.S. Heffner et al. 1994), and some of the larger mammals that have been tested have sucked the water instead of licking it (e.g., H.E. Heffner and R.S. Heffner 1986). These factors might keep the drinking noise at such a low level that the masking effect has been avoided.

In our experiment, the maximum level of the drinking noise far exceeded the auditory thresholds that have been determined for the pure tones (Tables 1, 2). Calculating the so-called spectrum level of the noise, which ideally is the intensity level of noise of a bandwidth of 1 Hz, one can estimate the critical ratio (CR), defined as the ratio of the level of the tone signal at the masked threshold for detection (L_{t+N_0}) to the spectrum level of the noise masker (L_{N_0}), i.e., $CR = L_{t+N_0} - L_{N_0}$. In doing so, we found a critical ratio for reindeer of -15 dB at 4 kHz, compared to a critical ratio of between 20 dB and 30 dB for most frequencies in man (Hawkins and Stevens 1950) and cats *Felis sylvestris catus* (Costalupes 1983). The explanation for this great difference cannot be that the reindeer ear is more acute in discrimination, but is probably a consequence of large variation in the drinking noise level. The drinking noise varied greatly, going down to a minimum between licks. The reindeer were probably able to detect the signal in the short low-intensity periods.

In the range of best sensitivity, reindeer had thresholds similar to other ungulates (R.S. Heffner and H.E. Heffner 1983, 1990; Wollack 1963), with the pig having slightly higher thresholds, cattle and goat having somewhat lower thresholds, while horse and sheep had both higher and lower thresholds than reindeer at different frequencies in the best sensitivity range (Fig. 2). This indicates a relatively low level of masking in this experiment, despite the high intensity of the drinking noise. However, the range of best sensitivity in reindeer was rather flat from 1 kHz to 16 kHz, a possible consequence when masking occurs at the frequencies with lowest thresholds. The phenomenon of a flat

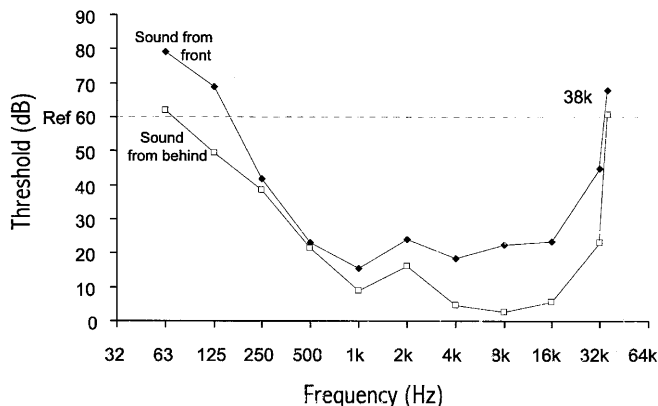


Fig. 1 Average audiogram of the two reindeer with sound source in front (filled symbols) and behind (open symbols). Ref 60 indicates the reference level of the hearing range. Thresholds in dB re. 20 μ Pa

Table 2 Individual hearing thresholds with sound from the front and sound from behind the animal

Frequency (Hz)	Reindeer 1, sound from front (dB)	Reindeer 2, sound from front (dB)	Reindeer 1, sound from behind (dB)	Reindeer 2, sound from behind (dB)
63	79	79	61	63
125	68	70	50	49
250	41	43	38	39
500	20	26	19	24
1 k	17	14	9	9
2 k	23	25	17	15
4 k	18	19	9	0
8 k	20	25	-1	6
16 k	24	23	7	4
32 k	48	42	25	21
38 k	—	68	63	59

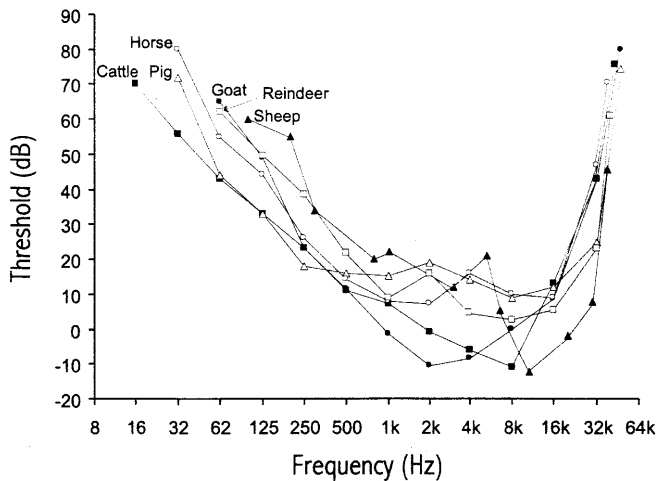


Fig. 2 Average audiogram of hoofed mammals. Thresholds in dB re. 20 μ Pa

best-frequency range is not uncommon in mammals. It has been observed in species such as horse, pig, lesser weasel (*Mustela nivalis*), cat, gerbil (*Meriones unguiculatus*), chinchilla (*Chinchilla laniger*), rabbit (*Oryctolagus cuniculus*), and kangaroo rat (*Dipodomys merriami*) (H.E. Heffner and Masterton 1980; R.S. Heffner and H.E. Heffner 1983, 1985a, 1985b, 1990, 1991; Ryan 1976). The flat best-frequency range in reindeer is therefore not likely to be an effect of masking.

It has been shown that the pinna is important in front-back discrimination, and in vertical localization of sound (R.S. Heffner and H.E. Heffner 1992; Musicant and Butler 1984). The size and shape of the pinna makes it effective in amplifying sound in its directional axis, and attenuating sound from other directions (Butler 1987; Calford and Pettigrew 1984; Carlile and Pettigrew 1987). In cats, the pinna is effective as a directional amplifier at frequencies higher than 2.5 kHz (Wiener et al. 1966). Since the reindeer had their pinnae directed backwards when drinking, the sound signals were played both from the front and from behind. As expected, lower thresholds were found at high frequencies with sound from behind, but unexpectedly lower thresholds were also found at 63 Hz and 125 Hz. The audiogram with sound from behind has lower thresholds than the audiogram with sound from the front on all frequencies. Reindeer are able to move their pinnae in the direction of a sound; therefore, the audiogram with sound from behind reflects the best audible sensitivity of reindeer.

The audiogram of domesticated ungulates such as cattle, horse, goat, pig, and sheep has been determined through behavioral tests (R.S. Heffner and H.E. Heffner 1983, 1990; Wollack 1963). Their hearing is similar to reindeer, with a mean of best sensitivity at 4.8 kHz (reindeer: 8 kHz), a mean low-frequency limit at 60 dB of 55 Hz (reindeer: 70 Hz), and a mean high-frequency limit at 60 dB of 37.5 kHz (reindeer: 38 kHz) (R.S. Heffner and H.E. Heffner 1990). Reindeer do not seem

to have any specialized hearing adaptations when comparing with these other ungulates.

Masterton et al. (1969) suggested that the ability to localize sound is the selective force behind high-frequency hearing ability in mammals, since high-frequency hearing is necessary for sound localization when the interaural distance is small. Confirming the theory, R.S. Heffner and H.E. Heffner (1990) showed a negative logarithmic relationship between maximum interaural distance and high-frequency hearing limit in mammals. As well as the other ungulates that have been tested, reindeer fit well into the linear relationship, with an interaural distance of 12 cm measured as the diameter of the head at the opening of the ear canal.

A good explanation for the relatively large variation in low-frequency hearing in mammals is lacking. Our results show that reindeer have higher hearing thresholds than other ungulates at most frequencies below 500 Hz, although goat and sheep have a somewhat higher low-frequency limit at the 60-dB level (R.S. Heffner and H.E. Heffner 1983, 1990; Wollack 1963).

Reindeer vocalization has fundamental frequencies of 1100 Hz (calve bleat), 630 Hz (female grunt), and even lower in rutting grunts by males (Lent 1972). These frequencies are much lower than the frequency range of best sensitivity in reindeer, but since attenuation with distance decrease with lower frequencies, the frequency level of vocalization might be optimal for detection at long distances.

Reindeer is the first cervid species to be tested for hearing ability. Cervids are key species in boreal forest and alpine and tundra ecosystems (Hobbs 1996), and are economically important for subsistence and sport hunting. Potential conflicts between sustainability of cervid populations and expanding human activities in wildlife habitats make it necessary to evaluate the resulting impact on animal behavior and area use. The reindeer audiogram makes it possible to assess the sensitivity of reindeer to different types of noise in connection with human activities.

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